

Insulin-like growth factor 1 (IGF1), IGF binding protein 3 (IGFBP3), and breast cancer risk: pooled individual data analysis of 17 prospective studies



Summary Background Insulin-like growth factor 1 (IGF1) stimulates mitosis and inhibits apoptosis. Some published results have shown an association between circulating IGF1 and breast-cancer risk, but it has been unclear whether this relationship is consistent or whether it is modified by IGF binding protein 3 (IGFBP3), menopausal status, oestrogen receptor status or other factors. The relationship of IGF1 (and IGFBP3) with breast-cancer risk factors is also unclear. The Endogenous Hormones and Breast Cancer Collaborative Group was established to analyse pooled individual data from prospective studies to increase the precision of the estimated associations of endogenous hormones with breast-cancer risk. **Methods** Individual data on prediagnostic IGF1 and IGFBP3 concentrations were obtained from 17 prospective studies in 12 countries. The associations of IGF1 with risk factors for breast cancer in controls were examined by calculating geometric mean concentrations in categories of these factors. The odds ratios (ORs) with 95% CIs of breast cancer associated with increasing IGF1 concentrations were estimated by conditional logistic regression in 4790 cases and 9428 matched controls, with stratification by study, age at baseline, and date of baseline. All statistical tests were two-sided, and a p value of less than 0.05 was considered significant. **Findings** IGF1 concentrations, adjusted for age, were positively associated with height and age at first pregnancy, inversely associated with age at menarche and years since menopause, and were higher in moderately overweight women and moderate alcohol consumers than in other women. The OR for breast cancer for women in the highest versus the lowest fifth of IGF1 concentration was 1.28 (95% CI 1.14–1.44; $p < 0.0001$). This association was not altered by adjusting for IGFBP3, and did not vary significantly by menopausal status at blood collection. The ORs for a difference in IGF1 concentration between the highest and lowest fifth were 1.38 (95% CI 1.14–1.68) for oestrogen-receptor-positive tumours and 0.80 (0.57–1.13) for oestrogen-receptor-negative tumours (p for heterogeneity=0.007). **Interpretation** Circulating IGF1 is positively associated with breast-cancer risk. The association is not substantially modified by IGFBP3, and does not differ markedly by menopausal status, but seems to be confined to oestrogen-receptor-positive tumours. **Funding** Cancer Research UK.